

Student Name: Harshika Bhosale

Roll No: #49784

Division / Batch: AUG26-PS102

THM Username :bhosaleharshika

3.1 Introduction

1.Computer networking means connecting two or more devices so that they can communicate and share information. Devices such as computers, mobile phones, servers, and printers can be connected using Wi-Fi or cables. The Internet is also a very large network that connects millions of devices around the world.

2.Networking is important in cybersecurity because most devices communicate through networks. A cybersecurity professional needs to understand how data moves from one device to another. Basic networking knowledge helps in finding suspicious network activity, understanding attacks, checking connections, and protecting systems from unauthorized access.

The two assigned TryHackMe rooms help in learning the basics of networking. They cover topics such as **networks, IP addresses, MAC addresses, packets, communication between devices, and basic network commands such as ping**. These concepts provide a basic foundation for understanding cybersecurity and network security.

Room Name-What is Networking?

3.2 What is Networking?

A **network** is a group of devices connected together so that they can communicate and share information.

For example, when your laptop and phone are connected to the same Wi-Fi, they are part of the same network.

Important Networking Terms

1. Network

A network connects devices so they can communicate with each other.

2. Client

A client is a device or program that **requests something**.

Example: Your browser is a client when you open a website.

3. Server

A server is a computer that **provides information or services** to clients.

Example: When you open YouTube, your device requests information from YouTube's servers.

4. Packet

When data is sent through a network, it is broken into small pieces called **packets**.

Think of it like sending a large parcel as several smaller packages.

5. IP Address

An IP address is an address used to identify a device on a network.

6. MAC Address

A MAC address is an address associated with a device's network interface, such as its Wi-Fi adapter.

7. Ping

Ping is a command used to check whether another device or server is reachable.

IP Address	MAC Address
IP stands for Internet Protocol	MAC stands for Media Access Control .
It is a logical address .	It is associated with the network interface .
It is used to identify a device on a network .	It is used to identify a network interface on the local network.
An IP address can change .	A MAC address is normally assigned to the network interface by the manufacturer.
Example: 192.168.1.10	Example: A4:C3:F0:85:AC:2D
IPv4 uses numbers separated by dots.	MAC addresses are usually written using hexadecimal characters.

Two computers communicate by **sending and receiving data through a network**. The data is divided into small pieces called **packets**.

For example, suppose **Computer A** wants to send data to **Computer B** on the same local network.

Computer A → Switch → Computer B

1. **Computer A wants to send data** to Computer B.
2. Computer A uses **IP addresses** to identify the destination.

3. For local communication, the device also needs the destination's **MAC address**.
4. The data is divided into **packets/frames**.
5. The **switch checks the destination MAC address** and forwards the data to the correct computer.
6. **Computer B receives the data** and processes it.
7. If Computer B needs to reply, it sends data back to Computer A in the same way.

Real-World Example

When you open **Google on your laptop**:

- Your laptop acts as the **client**.
- Google's computer acts as the **server**.
- Your laptop sends a request.
- The request travels through the network as **packets**.
- Google's server sends the requested information back.
- Your browser displays the webpage.

Questions solved in room

1)What is the key term for devices that are connected together?

->Network

Task 1 What is Networking?

Networks are simply things connected. For example, your friendship circle: you are all connected because of similar interests, hobbies, skills and sorts.

Networks can be found in all walks of life:

- A city's public transportation system
- Infrastructure such as the national power grid for electricity
- Meeting and greeting your neighbours
- Postal systems for sending letters and parcels

But more specifically, in computing, networking is the same idea, just dispersed to technological devices. Take your phone as an example; the reason that you have it is to access things. We'll cover how these devices communicate with each other and the rules that follow.

In computing, a network can be formed by anywhere from 2 devices to billions. These devices include everything from your laptop and phone to security cameras, traffic lights and even farming!

Networks are integrated into our everyday life. Be it gathering data for the weather, delivering electricity to homes or even determining who has the right of way at a road. Because networks are so embedded in the modern-day, networking is an essential concept to grasp in cybersecurity.

Take the diagram below as an example, Alice, Bob and Jim have formed their network! We'll come onto this a bit later on.



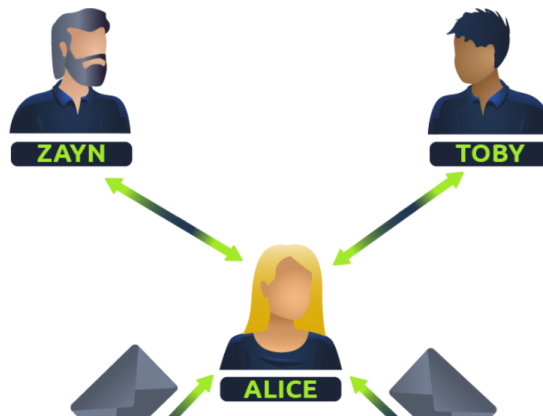
2) Who invented the World Wide Web?

-> Tim Berners-Lee

Task 2 What is the Internet?

Now that we've learnt what a network is and how one is defined in computing (just devices connected), let's explore the Internet.

The Internet is one giant network that consists of many, many small networks within itself. Using our example from the previous task, let's now imagine that Alice made some new friends named Zayn and Toby that she wants to introduce to Bob and Jim. The problem is that Alice is the only person who speaks the same language as Zayn and Toby. So Alice will have to be the messenger!



3) What does the term "IP" stand for?

->

Internet Protocol

4)

What is each section of an IP address called?

->

Octet

5)

How many sections (in digits) does an IPv4 address have?

->

4

6)

What does the term "MAC" stand for?

->

Media Access Control

7)

Deploy the interactive lab using the "View Site" button and spoof your MAC address to access the site. What is the flag?

->THM{YOU_GOT_ON_TRYHACKME}

Task 3 Identifying Devices on a Network

To communicate and maintain order, devices must be both identifying and identifiable on a network. What use is it if you don't know whom you're talking to at the end of the day?

Devices on a network are very similar to humans in the fact that we have two ways of being identified:

- Our Name
- Our Fingerprints

Now we can change our name through deed poll, but we can't, however, change our fingerprints. Every human has an individual set of fingerprints which means that even if they change their name, there is still an identity behind it. Devices have the same thing: two means of identification, with one being permeable. These are:

- An IP Address
- A Media Access Control (MAC) Address -- think of this as being similar to a serial number.

IP Addresses

Briefly, an IP address (or Internet Protocol) address can be used as a way of identifying a host on a network for a period of time, where that IP address can then be associated with another device without the IP address changing. First, let's split up precisely what an IP address is in the diagram below:

8)What protocol does ping use?

->

ICMP

9)

What is the syntax to ping 10.10.10.10?

->

ping 10.10.10.10

10)

What flag do you get when you ping 8.8.8.8?

->THM{I_PINGED_THE_SERVER}

Task 4 Ping (ICMP)

Ping is one of the most fundamental network tools available to us. Ping uses **ICMP** (Internet Control Message Protocol) packets to determine the performance of a connection between devices, for example, if the connection exists or is reliable.

[View Site](#)

The time taken for ICMP packets travelling between devices is measured by ping, such as in the screenshot below. This measuring is done using ICMP's echo packet and then ICMP's echo reply from the target device.

Pings can be performed against devices on a network, such as your home network or resources like websites. This tool can be easily used and comes installed on Operating Systems (OSs) such as Linux and Windows. The syntax to do a simple ping is `ping IP address or website URL`. Let's see this in action in the screenshot below.

```
cmnatic@CMNatic-THM-LPTOP:~$ ping 192.168.1.254
PING 192.168.1.254 (192.168.1.254) 56(84) bytes of data:
64 bytes from 192.168.1.254: icmp_seq=1 ttl=63 time=2.18 ms
64 bytes from 192.168.1.254: icmp_seq=2 ttl=63 time=2.53 ms
64 bytes from 192.168.1.254: icmp_seq=3 ttl=63 time=2.15 ms
64 bytes from 192.168.1.254: icmp_seq=4 ttl=63 time=3.34 ms
64 bytes from 192.168.1.254: icmp_seq=5 ttl=63 time=10.3 ms
64 bytes from 192.168.1.254: icmp_seq=6 ttl=63 time=4.45 ms
^C
--- 192.168.1.254 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 2.152/4.160/10.313/2.864 ms
cmnatic@CMNatic-THM-LPTOP:~$
```

Here we are pinging a device that has the private address of `192.168.1.254`. Ping informs us that we have sent six ICMP packets, all of which were received with an average time of 4.16 milliseconds.

Room name-Intro to Lan

- **3.3 Introduction to LAN**
- A **LAN (Local Area Network)** is a network that connects computers, phones, printers, servers, and other devices within a small geographical area such as a home, school, office, or college. The main purpose of a LAN is to allow connected devices to communicate and share resources such as files, printers, applications, and an Internet connection. LANs are usually faster and easier to manage because the devices are located within a limited area.
- A **switch** is an important device used in a LAN. It connects multiple devices together and helps send data to the correct device. A switch uses the **MAC addresses** of devices to decide where to forward data. For example, if Computer A wants to send data to Computer B, the switch checks Computer B's MAC address and forwards the data to the correct port instead of sending it to every device.
- Common components of a LAN include **computers, laptops, printers, servers, switches, routers, wireless access points, network cables, and network interface cards (NICs)**. Computers and laptops are used by users, while printers and servers provide shared resources. A switch connects devices within the LAN, while a router can connect the LAN to other networks such as the Internet. Wireless access points allow devices to connect using Wi-Fi.
- A **network topology** describes how devices in a network are physically or logically connected to each other. One common topology discussed in networking is the **star topology**. In a star topology, all devices are connected to a central device, usually a switch. For example, several computers in an office can each connect to one central switch. If one computer or its cable fails, the other computers can normally continue communicating. However, if the central switch fails, communication between the connected devices is affected.
- A real-world example of a LAN is a **college computer laboratory**. Many computers can be connected to a central switch, along with printers and other network devices. The

computers can communicate with each other, access shared resources, and connect to the Internet through a router. This allows the college to manage its devices and provide network access to students and staff.

Question asked in room

1)What does LAN stand for?

Local Area Network

2)What is the verb given to the job that Routers perform?

Routing

3)What device is used to centrally connect multiple devices on the local network and transmit data to the correct location?

Switch

4)What topology is cost-efficient to set up?

Bus Topology

5)What topology is expensive to set up and maintain?

Star Topology

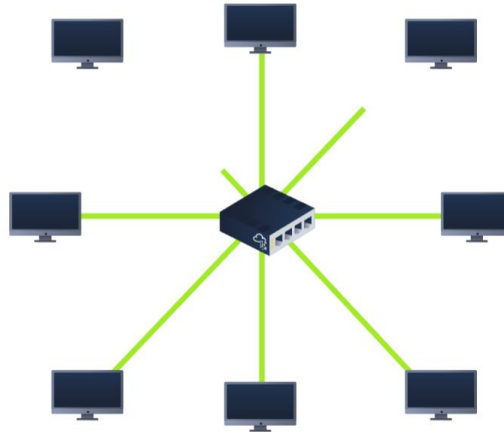
6)Complete the interactive lab attached to this task. What is the flag given at the end?

THM{TOPOLOGY_FLAWS}

Local Area Network (LAN) Topologies

[View Site](#)

Over the years, there has been experimentation and implementation of various network designs. In reference to networking, when we refer to the term "topology", we are actually referring to the design or look of the network at hand. Let's discuss the advantages and disadvantages of these topologies below.



7)What is the technical term for dividing a network up into smaller pieces?

Subnetting

8)How many **bits** are in a subnet mask?

32

9)What is the range of a section (octet) of a subnet mask?

0-255

10)What address is used to identify the start of a network?

Network Address

11)What address is used to identify devices within a network?

Host Address

12)What is the name used to identify the device responsible for sending data to another network?

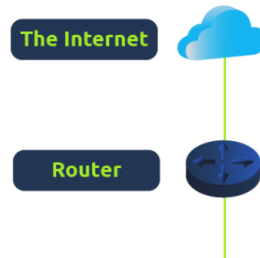
Default Gateway

Task 2 🟢 A Primer on Subnetting

As we've previously discussed throughout the module so far, Networks can be found in all shapes and sizes - ranging from small to large. Subnetting is the term given to splitting up a network into smaller, miniature networks within itself. Think of it as slicing up a cake for your friends. There's only a certain amount of cake to go around, but everybody wants a piece. Subnetting is you deciding who gets what slice & reserving such a slice of this metaphorical cake.

Take a business, for example; You will have different departments such as:

- Accounting
- Finance
- Human Resources



13) What does ARP stand for?

Address Resolution Protocol

14)

What category of ARP Packet asks a device whether or not it has a specific IP address?

Request

15)

What address is used as a physical identifier for a device on a network?

MAC Address

16)

What address is used as a logical identifier for a device on a network?

IP Address

Recalling from our previous tasks that devices can have two identifiers: A MAC address and an IP address, the **Address Resolution Protocol** or **ARP**, for short, is the technology that is responsible for allowing devices to identify themselves on a network.

Simply, ARP allows a device to associate its MAC address with an IP address on the network. Each device on a network will keep a log of the MAC addresses associated with other devices.

When devices wish to communicate with another, they will send a broadcast to the entire network searching for the specific device. Devices can use ARP to find the MAC address (and therefore the physical identifier) of a device for communication.

How does ARP Work?

Each device within a network has a ledger to store information on, which is called a cache. In the context of **ARP**, this cache stores the identifiers of other devices on the network.

In order to map these two identifiers together (IP address and MAC address), ARP sends two types of messages:

1. **ARP Request**
2. **ARP Reply**

When an **ARP request** is sent, a message is broadcasted on the network to other devices asking, "What is the mac address that owns this IP address?" When the other devices receive that message, they will only respond if they own that IP address and will send an **ARP reply** with its MAC address. The requesting device can now remember this mapping and store it in its **ARP cache** for future use.

17) What type of DHCP packet is used by a device to **retrieve an IP address**?

DHCP Discover

18)

What type of DHCP packet does a device **send once it has been offered an IP address** by the DHCP server?

DHCP Request

19)

Finally, **what is the last** DHCP packet that is sent to a device from a DHCP server?

DHCP ACK

IP addresses can be assigned either manually, by entering them physically into a device, or automatically and most commonly by using a **DHCP** (Dynamic Host Configuration Protocol) server. When a device connects to a network, if it has not already been manually assigned an IP address, it sends out a request (DHCP Discover) to see if any DHCP servers are on the network. The DHCP server then replies back with an IP address the device could use (DHCP Offer). The device then sends a reply confirming it wants the offered IP Address (DHCP Request), and then lastly, the DHCP server sends a reply acknowledging this has been completed, and the device can start using the IP Address (DHCP ACK).

